



Safety Data Sheet dated 27/2/2019, version 2

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product Identifier

Identification of the prepareate:

Trade name: DOMARK 40 EW

Trade code: -

1.2 Relevant identified uses of the substance/mixture and uses advised against

Agricultural use

Other use not admitted

1.3 Details of the supplier of the safety data sheet

Company:

ISAGRO S.p.A. – Via Caldera, 21 – 20153 – Milan - Italy

Emergency telephone number of the company and/or of an authorised advisory centre:

Tel.: 02 40 901 276

Competent person responsible for the safety data sheet:

msds@isagro.com

1.4 Emergency telephone number

QHSE Department (office time, local: 9.00-18.00) - Phone n.. ++39 02 40901209

SECTION 2: Hazards identification

2.1. Classification of the substance or mixture

In compliance with EC Regulation n. 1272/2008 (CLP):

 Aquatic Chronic 2, Toxic to aquatic life with long lasting effects.

Adverse physicochemical, human health and environmental effects:

No other hazards

2.2. Label elements

Hazard pictograms:



Hazard statements:

H411 Toxic to aquatic life with long lasting effects.

Precautionary statements:

P101 If medical advice is needed, have product container or label at hand.

P102 Keep out of reach of children.

P270 Do not eat, drink or smoke when using this product.

P280 Wear protective gloves/protective clothing/eye protection/face protection.

P391 Collect spillage.

P501 Dispose of contents/container in accordance with applicable regulations.

Special Provisions:

EUH401 To avoid risks to human health and the environment, comply with the instructions for use.

Special provisions according to Annex XVII of REACH and subsequent amendments:

None

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2.3. Other hazards
No vPvB and PBT in the mixture

Other Hazards:
No other hazards

SECTION 3: Composition/information on ingredients

3.1. Substances
Not applicable

3.2. Mixtures
Hazardous components within the meaning of the CLP regulation and related classification:

Qtà	Nome	Numero d'identif.	Classificazione
>= 3% - < 5%	(+/-) 2-(2,4-dichlorophenyl)- 3-(1 H -1,2,4-triazol-1-yl)propy l-1,1,2,2-tetrafluoroethy lether; tetraconazole	Numero 613-174-00-3 Index: CAS: 112281-77-3 EC: 407-760-7	 4.1/C2 Aquatic Chronic 2 H411  3.1/4/Oral Acute Tox. 4 H302  3.1/4/Inhal Acute Tox. 4 H332
>= 1% - < 3%	Di-(2-ethyl hexyl) sodium sulfosuccinate	CAS: 577-11-7 Reach: 01-21194912 96-29	 3.2/2 Skin Irrit. 2 H315  3.3/1 Eye Dam. 1 H318

SECTION 4: First aid measures

4.1. Description of first aid measures
In case of skin contact:
Wash with plenty of water and soap.
In case of eyes contact:
In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
In case of Ingestion:
Do not under any circumstances induce vomiting. OBTAIN A MEDICAL EXAMINATION IMMEDIATELY.
In case of Inhalation:
Remove casualty to fresh air and keep warm and at rest.
4.2. Most important symptoms and effects, both acute and delayed
None
4.3. Indication of any immediate medical attention and special treatment needed
Treatment:
None

SECTION 5: Firefighting measures

5.1. Extinguishing media
Suitable extinguishing media:
Water.
Carbon dioxide (CO2).
Extinguishing media which must not be used for safety reasons:
None in particular.
5.2 Special hazards arising from the substance or mixture
Do not inhale explosion and combustion gases which, at high temperatures, may contain toxic substances such as COx, NOx, HCl, HF and HCN.
Burning produces heavy smoke.
5.3 Advice for fire-fighters
Use suitable breathing apparatus. Self-contained breathing apparatus.

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Collect contaminated fire extinguishing water separately. This must not be discharged into drains.

Move undamaged containers from immediate hazard area if it can be done safely.

SECTION 6: Accidental release measures

- 6.1. Personal precautions, protective equipment and emergency procedures
 - Wear personal protection equipment.
 - Remove all sources of ignition.
 - Remove persons to safety.
 - See protective measures under point 7 and 8.
- 6.2. Environmental precautions
 - Do not allow to enter into soil/subsoil. Do not allow to enter into surface water or drains.
 - Retain contaminated washing water and dispose it.
 - In case of gas escape or of entry into waterways, soil or drains, inform the responsible authorities.
 - Suitable material for taking up: absorbing material, organic, sand
- 6.3. Methods and material for containment and cleaning up
 - Wash with plenty of water.
- 6.4. Reference to other sections
 - See also section 8 and 13

SECTION 7: Handling and storage

- 7.1. Precautions for safe handling
 - Avoid contact with skin and eyes, inhalation of vapours and mists.
 - Don't use empty container before they have been cleaned.
 - Before making transfer operations, assure that there aren't any incompatible material residuals in the containers.
 - Contaminated clothing should be changed before entering eating areas.
 - Do not eat or drink while working.
 - See also section 8 for recommended protective equipment.
- 7.2. Conditions for safe storage, including any incompatibilities
 - Keep away from food, drink and feed.
 - Incompatible materials:
 - None in particular.
 - Instructions as regards storage premises:
 - Adequately ventilated premises.
- 7.3. Specific end use(s)
 - None in particular

SECTION 8: Exposure controls/personal protection

- 8.1. Control parameters
 - No occupational exposure limit available
 - DNEL Exposure Limit Values
 - N.A.
 - PNEC Exposure Limit Values
 - N.A.
- 8.2. Exposure controls
 - Eye protection:
 - Use close fitting safety goggles, don't use eye lens.
 - Protection for skin:
 - Use clothing that provides comprehensive protection to the skin, e.g. cotton, rubber, PVC or viton.
 - Protection for hands:

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Use protective gloves that provides comprehensive protection, e.g. P.V.C., neoprene or rubber.

Respiratory protection:

Not needed for normal use.

Thermal Hazards:

None

Environmental exposure controls:

None

Appropriate engineering controls:

None

SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Properties	Value	Method:	Notes:
Appearance and colour:	Yellowish, liquid	--	--
Odour:	slight, characteristic	--	--
Odour threshold:	Not determined	--	--
pH:	6.25 (1% in water)	--	--
Melting point / freezing point:	< 0° C	--	--
Initial boiling point and boiling range:	> 100 °C	--	--
Flash point:	> 100.5 °C ° C	--	--
Evaporation rate:	Not relevant	--	--
Solid/gas flammability:	Not applicable	--	--
Upper/lower flammability or explosive limits:	Not determined	--	--
Vapour pressure:	0.14 mPa at 20 °C	--	Referred to <i>tetraconazole</i>
Vapour density:		--	--
Relative density:	1.058 at 20 °C kg/l	--	--
Solubility in water:	It gives emulsions	--	--
Solubility in oil:	Miscible in most common organic solvents	--	--
Partition coefficient (n-octanol/water):	Log P = 3.53	--	Referred to <i>tetraconazole</i>
Auto-ignition temperature:	450 °C	--	--
Decomposition temperature:	Not relevant	--	--
Viscosity:	Not relevant	--	--
Explosive properties:	Not explosive	--	On the basis of components
Oxidizing properties:	Not oxidizing	--	On the basis of components

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9.2. Other information

Properties	Value	Method:	Notes:
Miscibility:	Not available	--	--
Fat Solubility:	Not available	--	--
Conductivity:	Not available	--	--
Substance Groups relevant properties	Not available	--	--

SECTION 10: Stability and reactivity

- 10.1. Reactivity
Stable under normal conditions
- 10.2. Chemical stability
Stable under normal conditions
- 10.3. Possibility of hazardous reactions
None
- 10.4. Conditions to avoid
Stable under normal conditions.
- 10.5. Incompatible materials
None in particular.
- 10.6. Hazardous decomposition products
None.

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Data referred to the *mixture*:

Acute toxicity:

LD50 (oral): > 2000 (rat)

LD50 (dermale): > 2000 mg/kg

LC50 (4h) (inhalation): not available

Data referred to tetraconazole:

LC50 (4h) (inhalation) (OECD 403):

2.83 mg/l air (rat, male, nose only)

> 3.66 mg/l air (rat) (max attainable conc.)

Irritating power:

Skin: Not irritant (rabbit). Not classified as a skin irritating substance (OECD 404)

Eyes: Not irritant (rabbit). Not classified as a eye irritating substance (OECD 405)

Sensitization:

Skin: Not classified as sensitizer

(ECD 406, US EPA 81-6, Guinea Pig, Buehler Test; OECD 406, EEC B.6 – Guinea Pig

Short-term toxicity:

NOAEL oral:

rat, 90 gg.: 4.1 mg/kg bw/d (OECD 408; US EPA 82-1)

NOAEL oral:

dog, 1 year: 2.95 mg/kg bw/d (OECD 452; US EPA 83-1)

Carcinogenic effect (OECD 451):

No carcinogenic effect

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Mutagenic effect (EPA-TSCA 793400):
No mutagenic effect

Teratogenic effect (EPA-TSCA 793400):
No teratogenic effect

Reproduction toxicity (OECD 416):
No evidence of reproduction toxicity

SECTION 12. ECOLOGICAL INFORMATION

12.1 Toxicity

Adopt good working practices, so that the product is not released into the environment.

Data referred to the *mixture*:

Fish-

Zebra Fish (*Brachydanio rerio*), LC50 (96h):

> 100 mg/L

NOEC: 46 mg/L

Daphnia-

Daphnia magna, EC50 (48h): 42.1 mg/L

NOEC (48h): 4.6 mg/L

Algae-

Scenedesmus subspicatus,

Ec50: 10.4 mg/L

ErC50: 39.7 mg/L

NOEC (72h): 1.0 mg/L

NOEC (72h): 2.1 mg/L

12.2 Persistence and degradability

Data referred to *tetraconazole*:

Stable to hydrolysis and not expected to be degraded by photolysis in water.

Not readily biodegradable

12.3 Bioaccumulative potential

Data referred to *tetraconazole*:

BCF = 35.7 (whole fish)

12.4 Mobility in soil

Data referred to *tetraconazole*:

Koc between 531 e 1922

12.5 Results of PBT and vPvB assessment

Not requested, no PBT and/or vPvB substances in the mixture.

12.6 Other adverse effects

None

SECTION 13. DISPOSAL CONSIDERATIONS

13.1 Waste treatment methods

Recover if possible. In so doing, comply with the local and national regulations currently in force.

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SECTION 14. TRANSPORT INFORMATION

- 14.1. UN number
ADR-UN number: 3082
IMDG-Un number: 3082
- 14.2. UN proper shipping name
ADR-Shipping Name: 3082
IMDG-Technical name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (tetraconazole)
- 14.3. Transport hazard class(es)
ADR-Class: 9
ADR-Label: 9
ADR – Hazard identification number: 90
IMDG-Class: 9
IMDG-Label: 9 + Marine Pollutant
- 14.4. Packing group
ADR-Packing Group: III
IMDG-Packing group: III
- 14.5. Environmental hazards
Marine pollutant: Marine pollutant
- 14.6. Special precautions for user
IMDG-EMS: F-A, S-F
Limited Quantity: 5 L
Tunnel Restriction Code: (E)
- 14.7. Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code
Environmental Pollutant:
Not requested

SECTION 15: Regulatory information

- 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture
Dir. 98/24/EC (Risks related to chemical agents at work)
Dir. 2000/39/EC (Occupational exposure limit values)
Regulation (EC) n. 1907/2006 (REACH)
Regulation (EC) n. 1272/2008 (CLP)
Regulation (EC) n. 790/2009 (ATP 1 CLP) and (EU) n. 758/2013
Regulation (EU) 2015/830
Regulation (EU) n. 286/2011 (ATP 2 CLP)
Regulation (EU) n. 618/2012 (ATP 3 CLP)
Regulation (EU) n. 487/2013 (ATP 4 CLP)
Regulation (EU) n. 944/2013 (ATP 5 CLP)
Regulation (EU) n. 605/2014 (ATP 6 CLP)
Regulation (EU) n. 1221/2015 (ATP 7 CLP)
Regulation (EU) n. 918/2016 (ATP 8 CLP)
Regulation (EU) n. 1176/2016 (ATP 9 CLP)

Restrictions related to the product or the substances contained according to Annex XVII Regulation (EC) 1907/2006 (REACH) and subsequent modifications:

Where applicable, refer to the following regulatory provisions :

- Directive 82/501/EEC ('Activities linked to risks of serious accidents') and subsequent amendments.
Regulation (EC) nr 648/2004 (detergents).
1999/13/EC (VOC directive)

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Provisions related to directive EU 2012/18 (Seveso III):
Not applicable

15.2. Chemical safety assessment
Not requested

SECTION 16: Other information

H-statements in section 3:

H411 Toxic to aquatic life with long lasting effects.
H302 Harmful if swallowed.
H332 Harmful if inhaled.
H315 Causes skin irritation.
H318 Causes serious eye damage

Modified paragraphs with reference to the previous revision:

SECTION 2: Hazards identification
SECTION 3: Composition/information on ingredients
SECTION 14: Transport information
SECTION 15: Regulatory information
SECTION 16: Other information

This safety data sheet has been completely updated in compliance to Regulation 2015/830.

This document was prepared by a competent person who has received appropriate training.

Main bibliographic sources:

ECDIN - Environmental Chemicals Data and Information Network - Joint Research Centre,
Commission of the European Communities
SAX's DANGEROUS PROPERTIES OF INDUSTRIAL MATERIALS - Eight Edition - Van
Nostrand Reinold

The information contained herein is based on our state of knowledge at the above-specified date.
It refers solely to the product indicated and constitutes no guarantee of particular quality.
It is the duty of the user to ensure that this information is appropriate and complete with respect to
the specific use intended.

This MSDS cancels and replaces any preceding release.

ADR:	European Agreement concerning the International Carriage of Dangerous Goods by Road.
CAS:	Chemical Abstracts Service (division of the American Chemical Society).
CLP:	Classification, Labeling, Packaging.
DNEL:	Derived No Effect Level.
EINECS:	European Inventory of Existing Commercial Chemical Substances.
GefStoffVO:	Ordinance on Hazardous Substances, Germany.
GHS:	Globally Harmonized System of Classification and Labeling of Chemicals.
IATA:	International Air Transport Association.
IATA-DGR:	Dangerous Goods Regulation by the "International Air Transport Association" (IATA).
ICAO:	International Civil Aviation Organization.
ICAO-TI:	Technical Instructions by the "International Civil Aviation Organization" (ICAO).

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IMDG:	International Maritime Code for Dangerous Goods.
INCI:	International Nomenclature of Cosmetic Ingredients.
KSt:	Explosion coefficient.
LC50:	Lethal concentration, for 50 percent of test population.
LD50:	Lethal dose, for 50 percent of test population.
LTE:	Long-term exposure.
PNEC:	Predicted No Effect Concentration.
RID:	Regulation Concerning the International Transport of Dangerous Goods by Rail.
STE:	Short-term exposure.
STEL:	Short Term Exposure limit.
STOT:	Specific Target Organ Toxicity.
TLV:	Threshold Limiting Value.
TWATLV:	Threshold Limit Value for the Time Weighted Average 8 hour day. (ACGIH Standard).
WGK:	German Water Hazard Class.